SE311 SPRING 2025-2026 / 29-04-2026 & 30-04-2026
Lab Work 08 – Template Method Pattern

At **Secure Flow Banking Solutions**, processing financial transactions is a highly regulated and systematic operation. The system handles various transaction types, such as **Cash Withdrawals** and **International Transfers**.

Although these transactions involve different banking protocols, they all follow a strictly enforced execution pipeline:

1. **Verify Identity:** Authenticate the user through PIN, biometrics, or digital signatures.
2. **Check Balance:** Ensure the account has sufficient funds for the requested amount.
3. **Execute Transfer:** Move the funds from the source to the destination.
4. **Update Ledger:** Record the transaction in the bank's permanent database.
5. **Send Notification:** Notify the user via SMS or Email about the transaction status.

The output of the procedure above will look like this on the console:

[CashWithdrawal]
Verifying identity…
Checking available balance in savings account.
Dispensing cash from ATM.
Updating transaction ledger.
Sending notification to customer…

[InternationalTransfer]
Verifying identity…
Checking available balance in savings account.
Processing SWIFT transfer to the destination bank.
Updating transaction ledger.
Sending notification to customer…

Implement necessary classes and methods to simulate this scenario and implement a template method for that system.

After a recent security audit, the compliance team mandated that for **International Transfers**, a **Currency Compliance & Fraud Screening** step must be performed between Step 1 and Step 2 to satisfy anti-money laundering (AML) laws. This process is not required for standard local cash withdrawals. After that change, the output for the international transfer will look like this:

[InternationalTransfer]
Verifying identity…
Performing AML compliance check and fraud screening...
Checking available balance in savings account.
Processing SWIFT transfer to the destination bank.
Updating transaction ledger.
Sending notification to customer…

Adjust the template method for those changes.